Reverse Percentages

Exam code: 0607

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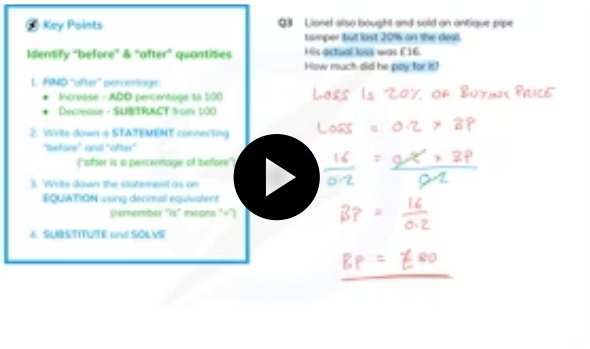
Updated on 4 February 2026



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Reverse percentages

What is a reverse percentage?

- A reverse percentage question is one where we are given the **value after a percentage increase or decrease** and asked to find the value **before** the change

Help me understand

Examiner Tips and Tricks

To spot a reverse percentage question, see if you are being asked to find a quantity in the past.

E.g. Find the **old** / **original** / **before** amount ...

How do I solve reverse percentage questions?

- You should think about the **before quantity**
  - even though it is not given in the question
- Find the percentage change as a **multiplier**, p
  - This is the decimal equivalent of a percentage change

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$= 0.95$

- Use **before × p = after** to write an **equation**
  - Get the **order** right: the percentage change happens to the "before", not to the "after"
- **Rearrange** the equation to make the "before" quantity the subject
  - **Divide** the "after" quantity by the multiplier, p
  - $\text{Before} = \dfrac{\text{After}}{\text{Multiplier}}$

Examiner Tips and Tricks

Avoid the **common mistake** of thinking that increasing an amount by a percentage is the inverse of decreasing an amount by that percentage.

For example, increasing 100 by 10% gives 110. But decreasing 110 by 10% gives 99 not the original 100.

Worked Example

Jennie has been working for a company for the last ten years.

She receives a pay rise of 5%.

Her new salary is £31 500 per year.

Find her salary before the pay rise.

Answer:

Use "before" × p = "after" to write an equation

The "before" amount is unknown and the "after" amount is 31 500

$$\text{"before"} \times 1.05 = 31\,500$$

Find the multiplier, p (by writing 5% as a decimal and adding it to 1)

$$p = 1 + 0.05 = 1.05$$

Find the value of "before" (by dividing both sides by 1.05)

$$\text{"before"} = \frac{31\,500}{1.05} = 30\,000$$

She was paid £30 000 before the pay rise

Jennie was paid £30 000 per year before the pay rise

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